Groundwater

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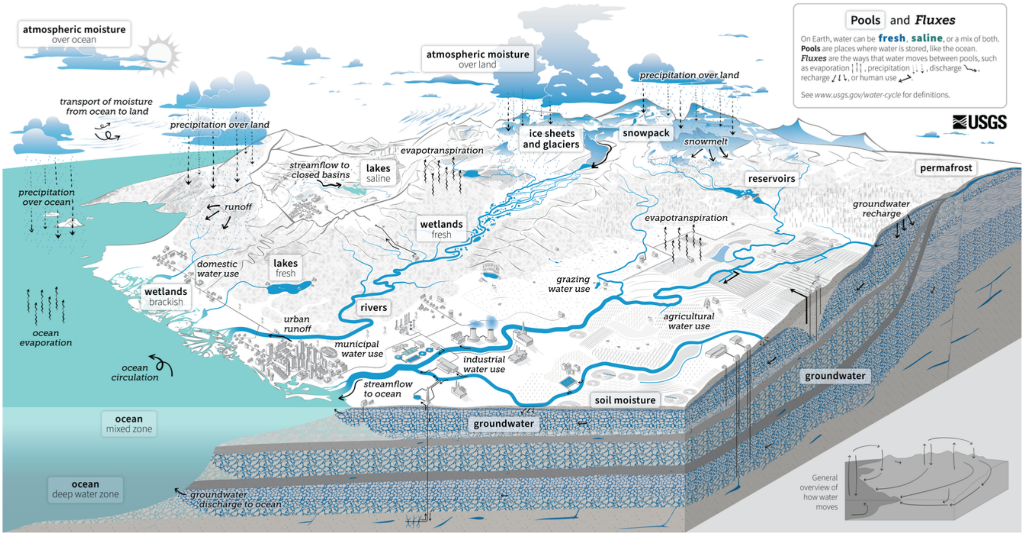
Housekeeping

- Exam re-take during lab today
- Exam redo (handwritten on clean sheets of paper) can be submitted at end of lab, or in person next Tuesday at start of class
- Show your work!
- PS#6 is live, due next Thursday at midnight

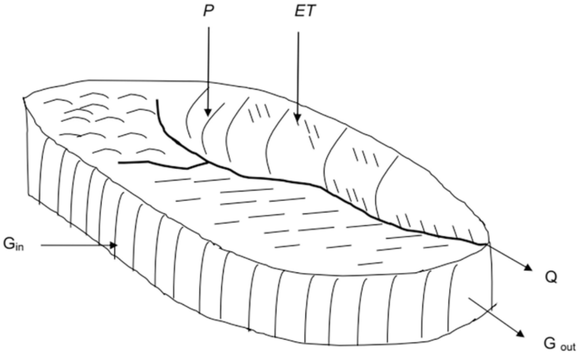
Water Cycle



atmospheric moisture
over ocean
transport of moisture from ocean to land
precipitation over land
atmospheric moisture
over land
precipitation over land
Pools and Fluxes
On Earth, water can be fresh, saline, or a mix of both.
Pools are places where water is stored, like the ocean.
Fluxes are the ways that water moves between pools, such as evaporation, precipitation, discharge, recharge, or human use.
See www.usgs.gov/water-cycle for definitions.
USGS
precipitation over ocean
runoff
streamflow to closed basins
lakes
saline
evapotranspiration
ice sheets and glaciers
snowpack
snowmelt
reservoirs
permafrost
wetlands
fresh
evapotranspiration
groundwater recharge
domestic water use
lakes
fresh
wetlands
brackish
ocean evaporation
rivers
grazing water use
agricultural water use
urban runoff
municipal water use
industrial water use
ocean circulation
streamflow to ocean
soil moisture
groundwater
groundwater
ocean
mixed zone
ocean
deep water zone
groundwater discharge to ocean
General overview of how water moves

Regional Water Balance

$$\hat{P} + G_{in} - (Q + ET + G_{out}) = \Delta S$$



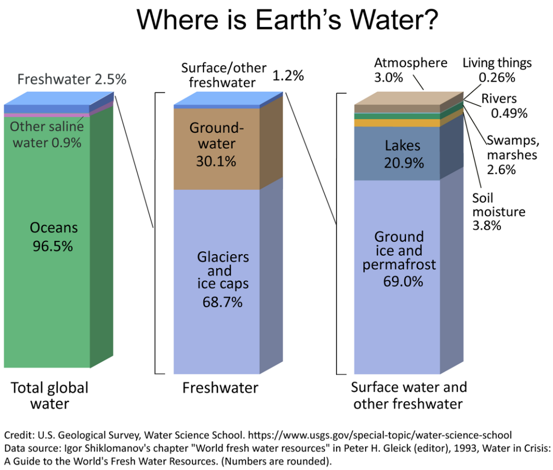
Where:

P = Precipitation
ET = Evapotranspiration
Q = Stream outflow
G_{in} = Ground water in
G_{out} = Ground water out

Learning Objectives

1. Aquifer Types and Properties
2. Groundwater Flow Equation(s)
3. Regional Groundwater Flow
4. GW/SW Interactions
5. Groundwater Balance Components
6. Groundwater Development

Importance of Groundwater



- Groundwater (GW) comprises 96% of liquid freshwater
- Major source of water for irrigation and drinking water globally
- Major source of water to streams and lakes
- In arid/semi-arid regions, it's a critical buffer for when surface water (SW) is not available.

Aquifers and Aquitards

Geologic formation: regionally extensive, mappable geologic unit with a characteristic lithology.

Aquifer: A formation, a group of formations, or a part of a formation that contains sufficient saturated permeable material to yield significant quantities of water to wells and springs. (USGS)

Aquitard: Formations that do not transmit water at hydrologically significant rates, i.e., impermeable.

Aquifer and aquitard are relative terms with different meaning to different practitioners.

Principal Aquifers

Major Aquifers in U.S.

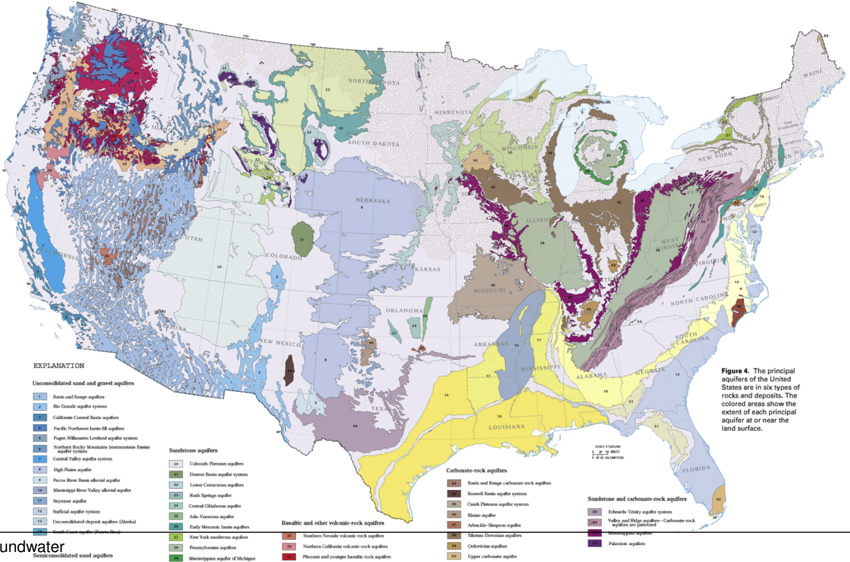
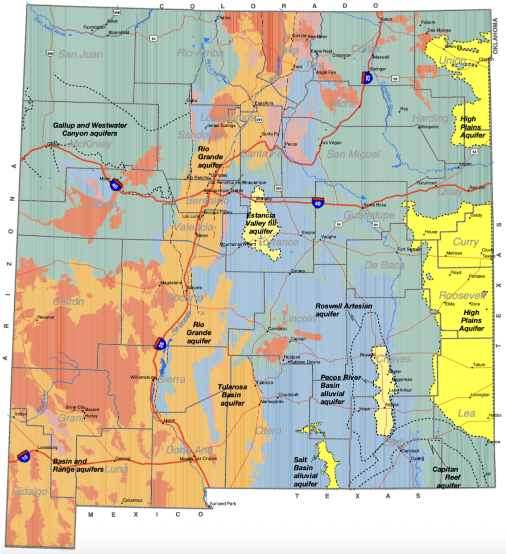


Figure 4. The principal aquifers of the United States are in six types of rocks and deposits. The colored areas show the extent of each principal aquifer at or near the land surface.

New Mexico Aquifers



Explanation:

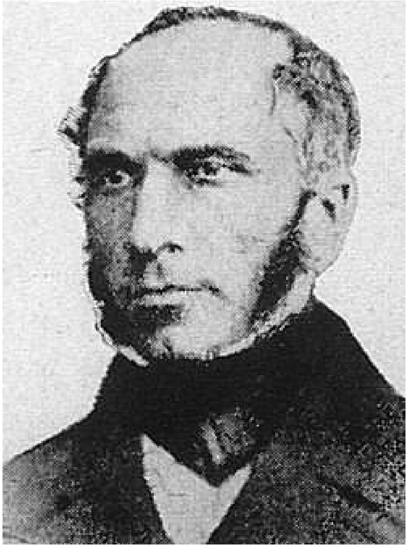
- Thin alluvial-fan and river-laid deposits (important aquifers)
- Basin fill in deep down-faulted basins (important aquifers)
- Volcanic rocks (not major aquifers)
- Sandstone and shale aquifers (locally significant aquifers)
- Limestone, sandstone, and shale aquifers (local aquifers)
- Granite and other similar crystaline rocks (not major aquifers)
- Major faults

Darcy's Law

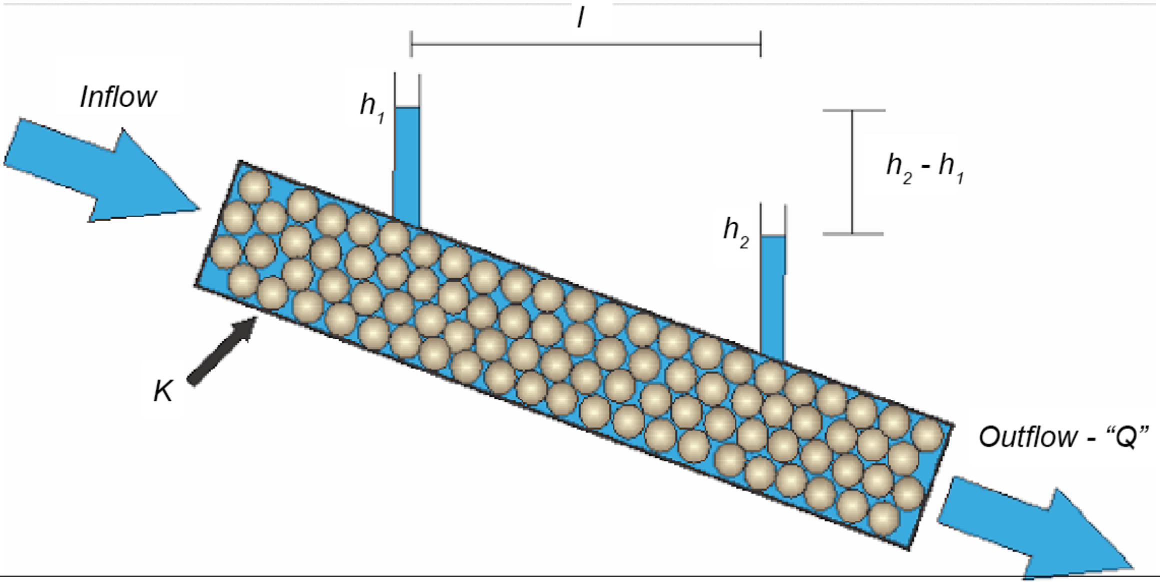
$$\frac{Q}{A} = q = K \cdot \frac{dh}{dl}$$

where q is the specific discharge or Darcy

flux (L/T), Q is volumetric flux (L^3/T), A is cross-sectional area (L^2), K is hydraulic conductivity, l is length, and h is **hydraulic head** (h).



Darcy Tube

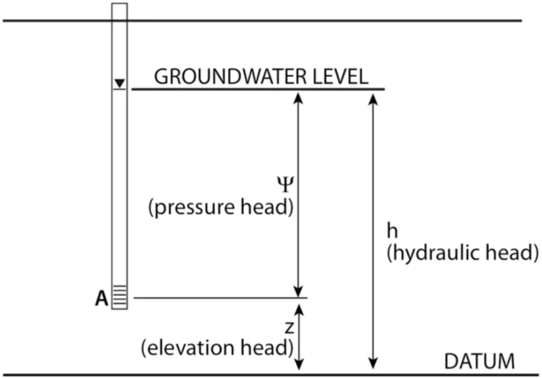


Hydraulic Head

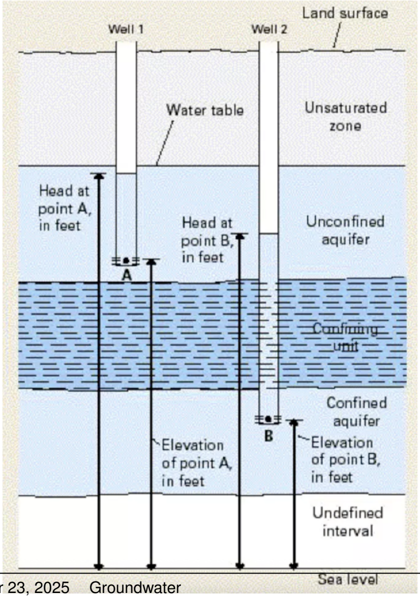
Water moves through the subsurface as a result of gravitational force and pressure gradients. Assuming the density of water is constant, the sum of these drivers is the **hydraulic head**, h, with units L.

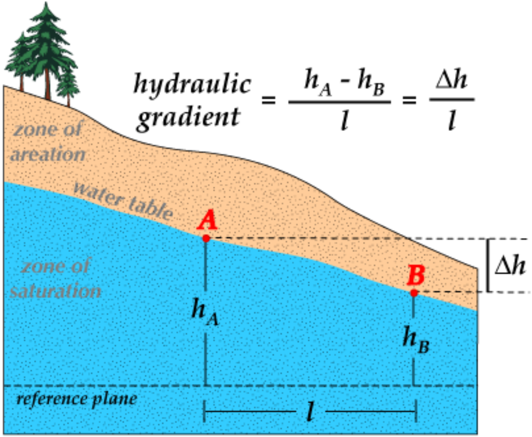
$$h = z + \frac{p}{\rho_w \cdot g}$$

where z is elevation above a datum, p is pressure, ρ_w is density of water, and g is gravitational acceleration.



Hydraulic Head

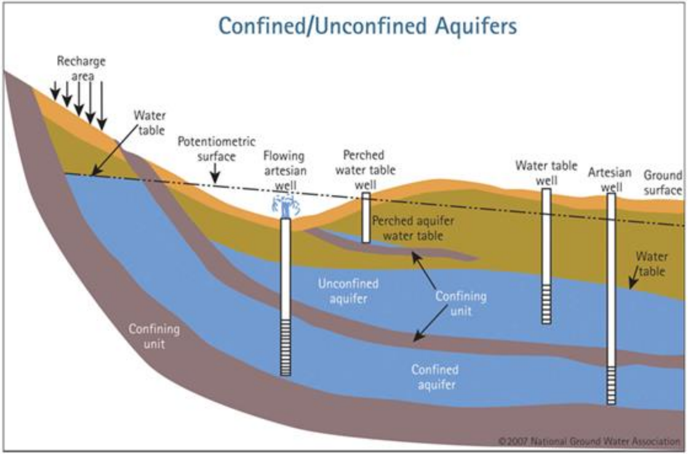




h = hydraulic head

Unconfined Aquifer

Upper boundary of saturated zone (water table) is open to the atmosphere. Recharge occurs as infiltrating water percolating downwards over a large area. Flow and storage are correlated.



Wait, what's recharge?

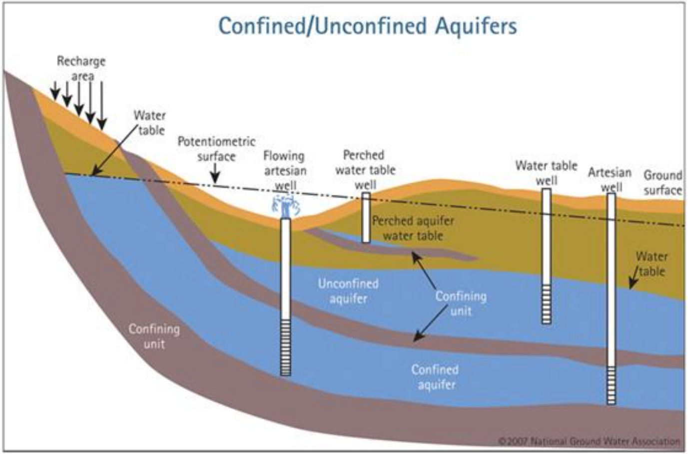
Water that enters the saturated zone via percolation of water through the vadose zone or as vertical/horizontal seepage from surface water bodies. It is how aquifers are **replenished**.

Under natural conditions, groundwater recharge is typically balanced by groundwater discharge to rivers, lakes, and oceans such that storage does not substantially change.

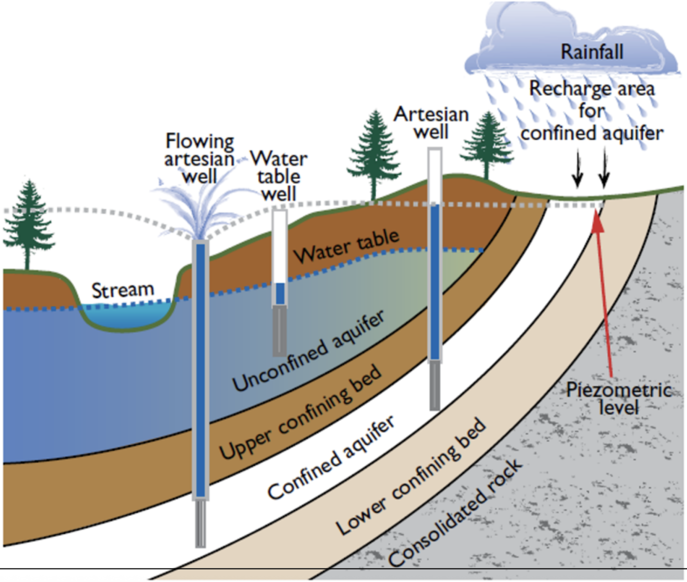
This is a **steady-state** condition.

Confined Aquifer

When the upper boundary of aquifer is an aquitard, it is **confined.** The pressure throughout is greater than atmospheric–water in a well penetrating the confined aquifer will rise above the confining unit. This is referred to as the **potentiometric/piezometric surface.**

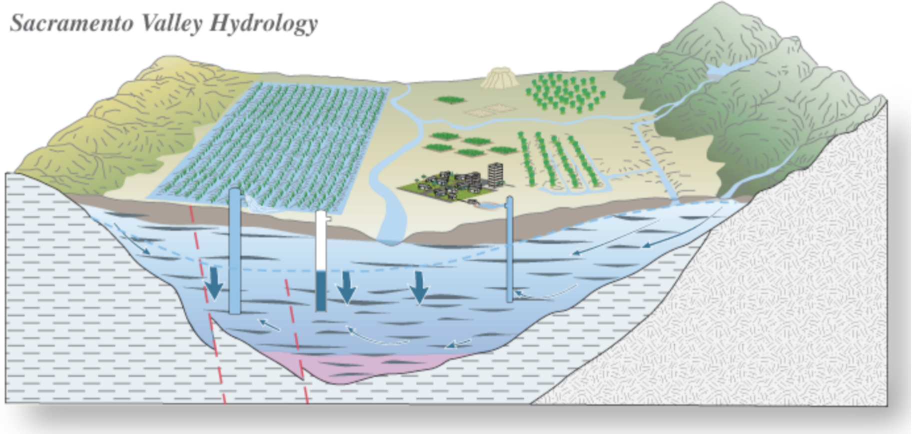


Artesian Wells



Semi-confined Aquifer

No porous medium is truly impermeable, and some are permeable enough the be termed **leaky aquitards** that transmit water from overlying aquifers. In other cases, the confining units are discontinuous, allowing direct connection to overlying units.



Storage Properties of Unconfined Aquifers

Change in water table produces a change in volume of water stored in the porous medium. The amount of water content change is characterized by the **specific yield** (S_y), or volume released per unit surface area per unit decline of water table.

As the water table declines, some water is retained in the newly unsaturated void space, hence:

$$\phi = S_y + S_r$$

where S_r is the volume of water retained per unit decline in water table per unit surface area. What vadose zone term is this related to?

Storage Properties of Confined Aquifers

Confined aquifers have a constant thickness (b) that is unaffected by storage changes. Instead, storage changes due to:

- compaction of the aquifer skeleton
- expansion of the water due to the lowered pressure

The volume of water released per unit change in head in a confined aquifer is characterized by **specific storage** (S_s),

$$S_s = (\alpha + \phi \cdot \beta) \cdot \gamma_w$$

where α is the compressibility of the aquifer, β is the compressibility of water, ϕ is the porosity, and γ_w is the weight density of water.

Storage Properties of Confined Aquifers

$$S_s = (\alpha + \phi \cdot \beta) \cdot \gamma_w$$

At near-surface pressures and temperatures, β is very small and relatively constant (4.4x10^{-10} m^2/N).

Confined storage is often expressed as the dimensionless **storativity**.

$$S = S_s \cdot b$$

where b is aquifer thickness (L).

Material	Compressibility, α (m^2/N or Pa^{-1})
Clay	10^{-8} to 10^{-6}
Sand	10^{-9} to 10^{-7}
Gravel	10^{-10} to 10^{-8}
Jointed rock	10^{-10} to 10^{-8}
Sound rock	10^{-11} to 10^{-9}

Transmission Properties of Aquifers

Hydraulic conductivity (K), units of (L/T) is the volume rate of flow per unit area transmitted through the porous medium under a unit hydraulic gradient. K is spatially variable and direction-dependent in aquifers.

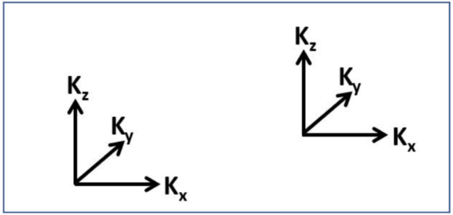
Isotropy: K is the same in all (x,y,z) directions. K that differs depending on direction is **anisotropic**. K_v typically < K_h (commonly 10%).

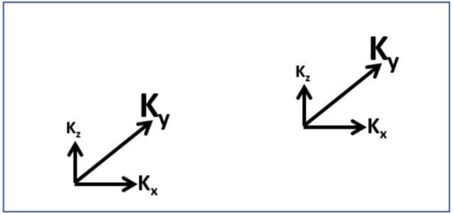
Homogeneous: K is constant in space. K varying in space is **heterogeneous**.

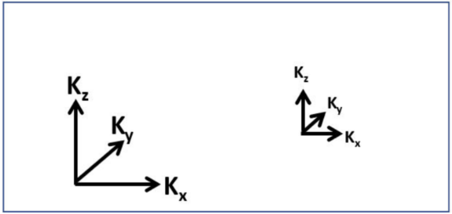
Transmissivity (units 1/T) describes the aquifer's overall capacity to transmit water.

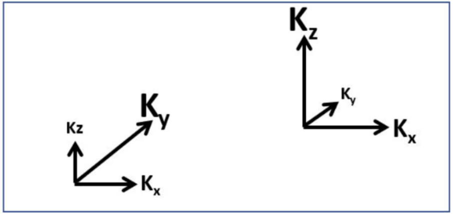
$$T = K \cdot b \tag{1}$$

Hydraulic Conductivity

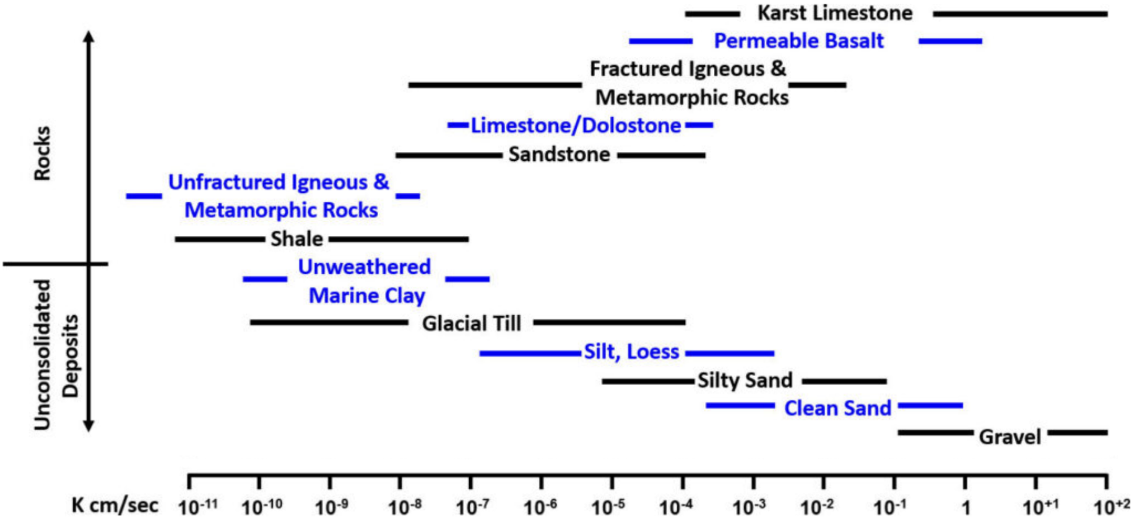








Hydraulic Conductivity



Estimating K

- Permeameters
- Empirical Relationships
- Hydraulic Testing

Permeameters

Can be performed in the lab or in the field. Guelph permeameter is a field instrument that maintains a constant-head to estimate saturated K.

Limited to first 1-3 m of subsurface.

What is this similar to?

Empirical Relationships

Relate grain-size distribution to K. Only for unconsolidated sediments.

Hazen (1911) method is the simplest:

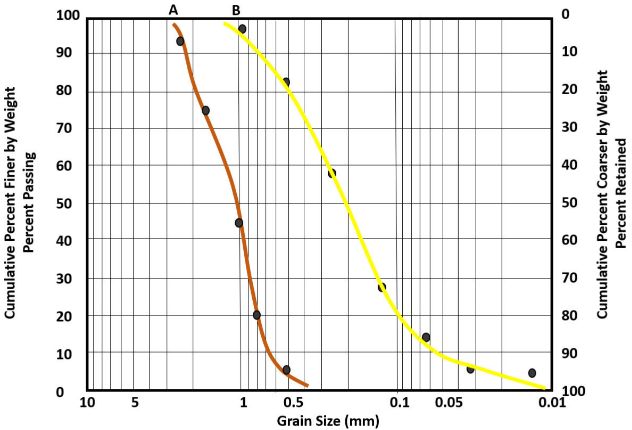
$$K = C \cdot d_{10}^2$$

where $d_1 0$ is the effective grain size in cm where 10% of the material is finer and 90% is retained and C is a coefficient based on the following:

Material	$\boldsymbol{C}$
Very fine sand, poorly sorted	40-80
Fine sand with appreciable fines	40-80
Medium sand, well sorted	80-120
Coarse sand, poorly sorted	80-120
Coarse sand, well sorted, clean	120-150

Hazen Method

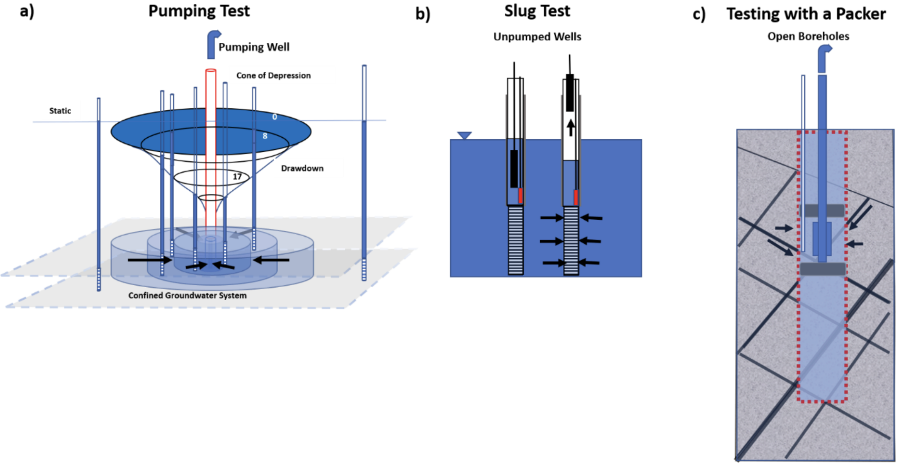
Which one is likely to have a higher K? What is the total sample K for sample A?



Hydraulic Testing

Determine transmissivity (and in some cases, storativity) at the field-scale using the following:

1. Pump test
2. Slug test
3. Packer test



Groundwater Flow Equation

3-D, transient, anisotropic, heterogeneous. Derived from Darcy's Law and conservation of mass:

$$S_s \frac{\partial h}{\partial t} = \frac{\partial}{\partial x} K_x (\frac{\partial h}{\partial x}) + \frac{\partial}{\partial y} K_y (\frac{\partial h}{\partial y}) + \frac{\partial}{\partial z} K_z (\frac{\partial h}{\partial z})$$

Homogeneous K

$$S_s \frac{\partial h}{\partial t} = \frac{\partial}{\partial x} K_x (\frac{\partial h}{\partial x}) + \frac{\partial}{\partial y} K_y (\frac{\partial h}{\partial y}) + \frac{\partial}{\partial z} K_z (\frac{\partial h}{\partial z})$$

Isotropic Flow

$$S_s \frac{\partial h}{\partial t} = \frac{\partial}{\partial x} K_x (\frac{\partial h}{\partial x}) + \frac{\partial}{\partial y} K_y (\frac{\partial h}{\partial y}) + \frac{\partial}{\partial z} K_z (\frac{\partial h}{\partial z})$$

Steady-state flow

$$S_s \frac{\partial h}{\partial t} = \frac{\partial}{\partial x} K_x (\frac{\partial h}{\partial x}) + \frac{\partial}{\partial y} K_y (\frac{\partial h}{\partial y}) + \frac{\partial}{\partial z} K_z (\frac{\partial h}{\partial z})$$

2-D, Unconfined, Transient flow

Dupuit assumption: flow is predominantly horizontal (and therefore 2-D), such that dh/dl = dh/dx, b=h, and flow in the vertical can be neglected.

$$S_y \frac{\partial h}{\partial t} = \frac{\partial}{\partial x} K_x (h \frac{\partial h}{\partial x}) + \frac{\partial}{\partial y} K_y (h \frac{\partial h}{\partial y})$$

- Homogeneous:

- Steady-state: